Intelligent Cyber Crime Reporting System

Batch Number: CCS_03

Roll Number	Student Name
20231CCS3001	ANKIREDDY BHARATH KUMAR REDDY
20221CCS0081	V S Sai Datta Yagnesh
20221CCS0069	Venkat Prasad

Under the Supervision of,

Ms.Soumya

Assistant Professor
School of Computer Science and Engineering
Presidency University

Name of the Program: **Computer Science And Engineering-CYBER SECURITY**
Name of the HoD: **Dr.Anandaraj**
Name of the Program Project Coordinator: **Dr. Sharmasth Vali Y**
Name of the School Project Coordinators: **Dr. Sampath A K , Dr. Geetha A**

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Problem Statement Number:SIH25132

Organization: AICTE.
Category :Software

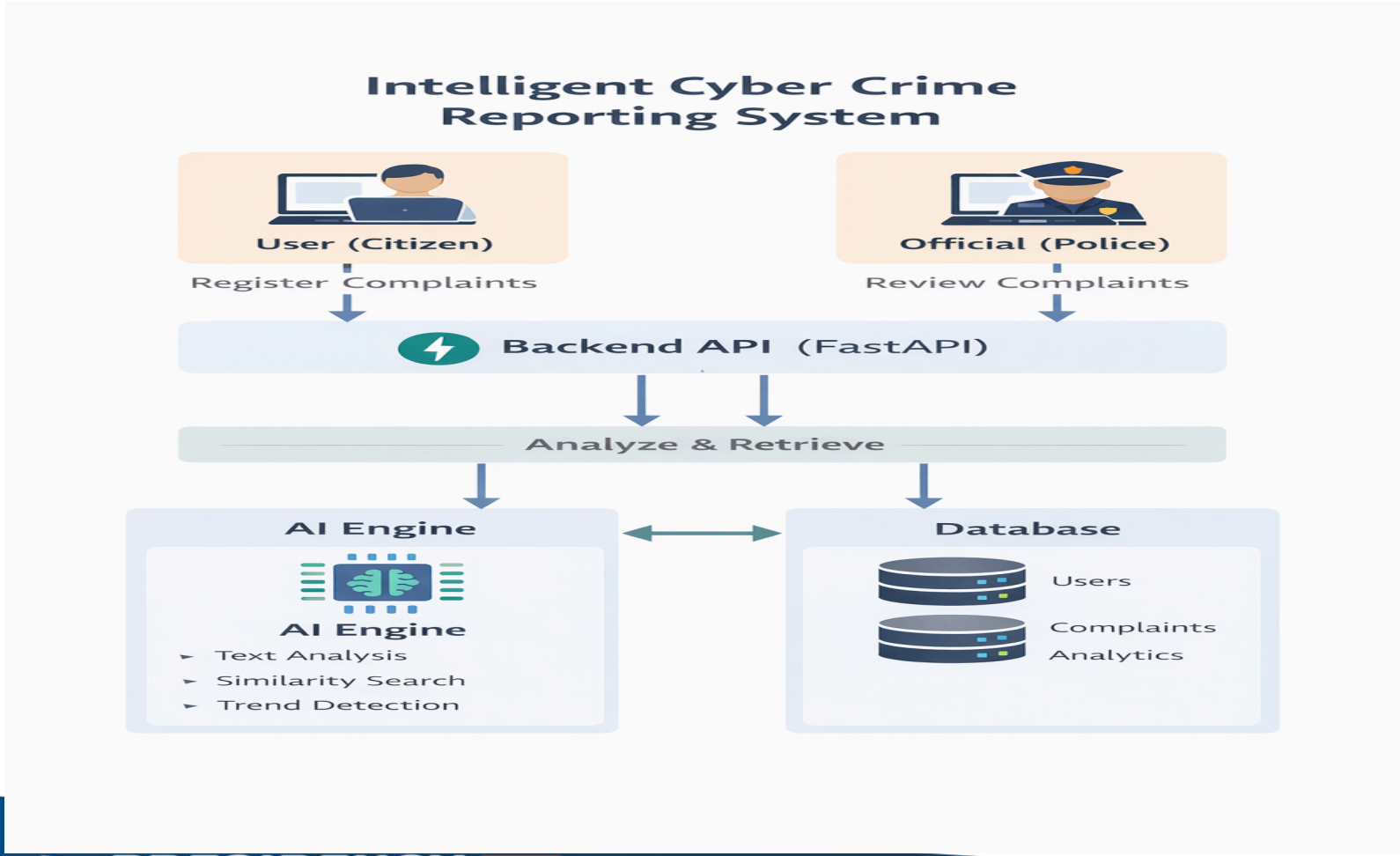
Problem Description:

- Cyber crimes such as online fraud, phishing, identity theft, and social engineering attacks are increasing rapidly in both volume and complexity. Although online cyber crime reporting portals exist, they primarily focus on complaint registration and storage. These systems lack intelligent mechanisms to analyze newly registered complaints, correlate them with previous similar cases, and assist law enforcement officials with insights such as crime patterns, modus operandi, and resolution history.
- As a result, investigators often spend significant time manually searching past records, leading to delayed investigations, repeated patterns of crime, and slower case resolution. Additionally, citizens lack access to real-time awareness dashboards that inform them about emerging cyber crime trends and newly evolving attack techniques.

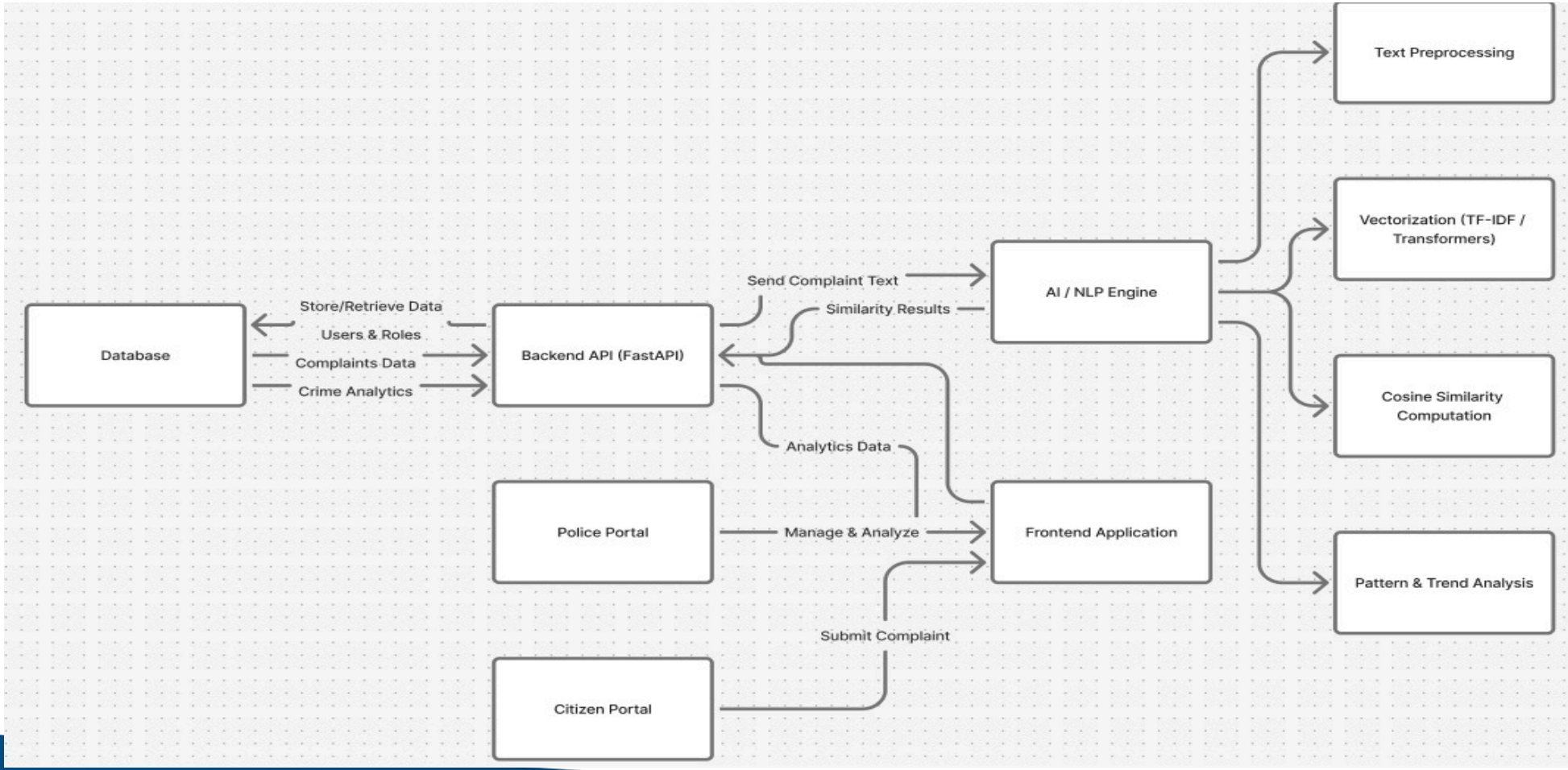
Objectives

- To design a centralized cyber crime reporting platform for citizens and officials.
- To implement AI and Machine Learning techniques for analyzing complaint descriptions.
- To identify and display similar past cyber crime cases using similarity scoring.
- To assist law enforcement by presenting modus operandi, crime patterns, and case resolution summaries from historical data.
- To provide citizens with an awareness dashboard showing new and emerging cyber crime trends.
- To reduce investigation time and improve decision-making efficiency for officials.
- To ensure data privacy, role-based access, and ethical use of AI.

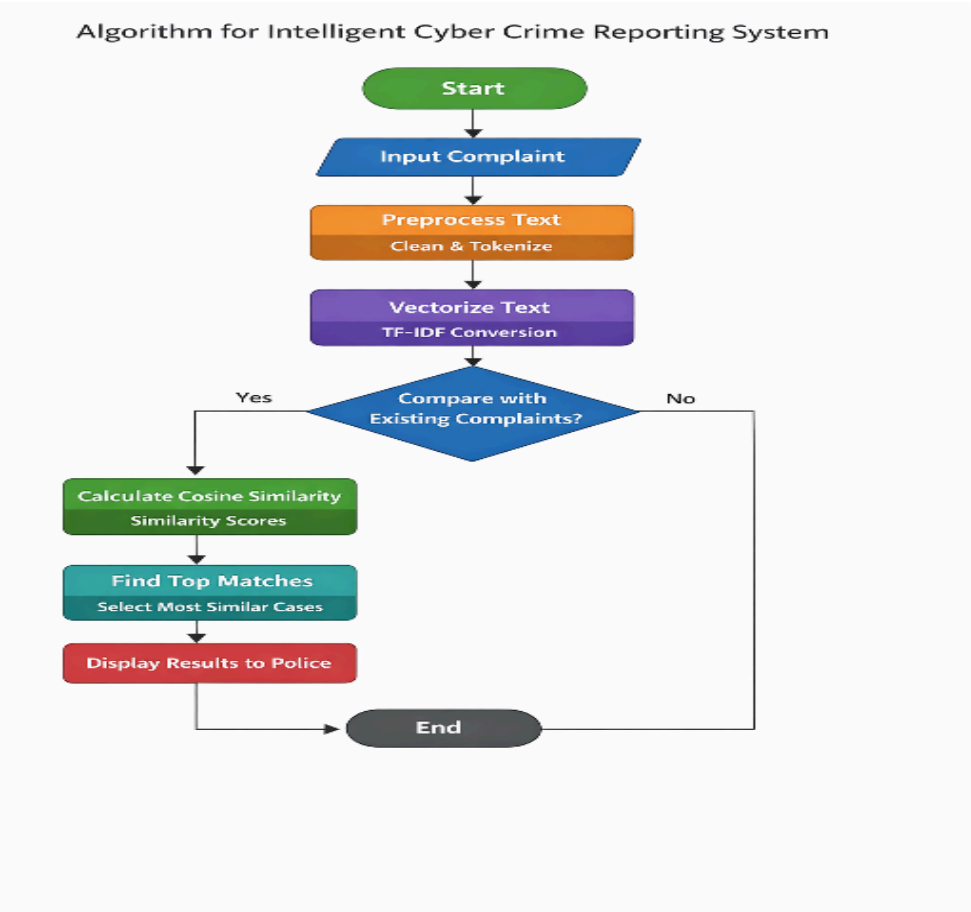
Architecture Diagram



BLOCK DIAGRAM



ALOGORITHM


Algorithm for Intelligent Cyber Crime Reporting System
Start
Input Complaint
Preprocess Text
Clean & Tokenize
Vectorize Text
TF-IDF Conversion
Compare with Existing Complaints?
Yes
No
Calculate Cosine Similarity
Similarity Scores
Find Top Matches
Select Most Similar Cases
Display Results to Police
End

Drawbacks of the System

- **Limited Understanding of Context**
- The system uses TF-IDF and cosine similarity, which mainly rely on keyword matching. It may fail to fully understand the actual meaning or context of complex complaints.

- **Dependence on Data Quality**
- The accuracy of results depends heavily on the quality of complaint data. Incomplete, unclear, or poorly written complaints can lead to incorrect similarity results.

- **Lack of Deep Learning Capabilities**
- The current system does not use advanced models like BERT or deep learning techniques, which could provide better semantic understanding and accuracy.

- **Scalability Challenges**
- As the number of complaints increases, computing similarity with all existing data may become slower and require optimization or more powerful infrastructure.

- **Limited to Text-Based Analysis**
- The system only analyzes textual data and does not consider multimedia evidence such as images, videos, or audio, which are important in many cyber crime cases.

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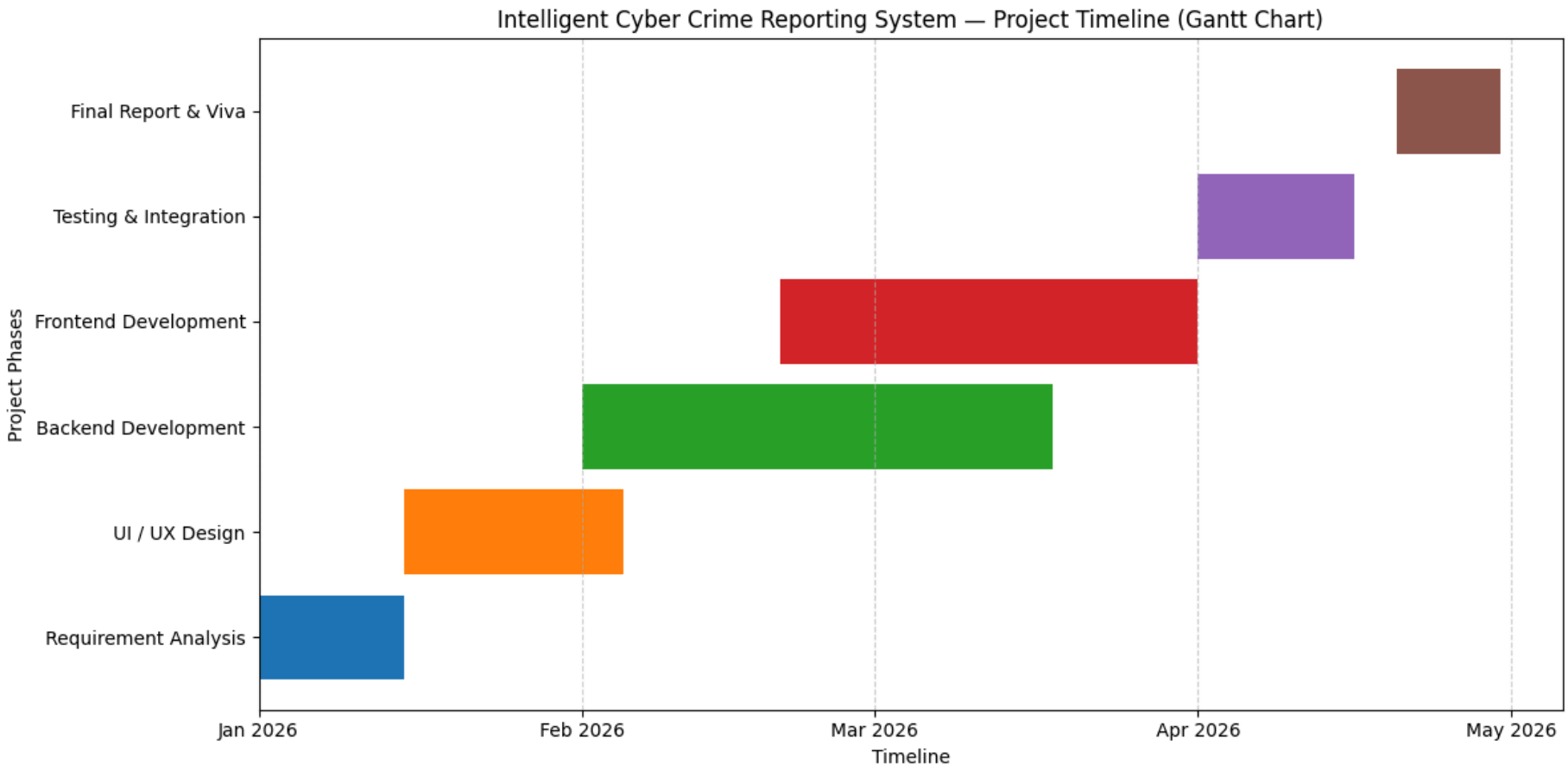
Github Link

The Github link provided should have public access permission.

Github Link

Timeline of the Project (Gantt Chart)



Thank
You!